

termsync



databricks



ROGERS

TRANS



LINK



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Why are we here?

Today, we aim to build a **smart community solution** in Databricks for Transit or Security using Roger's cell tower traffic data across Vancouver. We chose **Transit at UBC**, where the region's most crowded bus routes meet.

3

locations

UBC, Waterfront Station,
Park Royal

10

months

Nov 2025 - Aug 2026

21M+

connections

Timestamped cell sessions

2

key signals

Origin data and dwell time

**How many of you have
waited for more than one
R4 at UBC?**

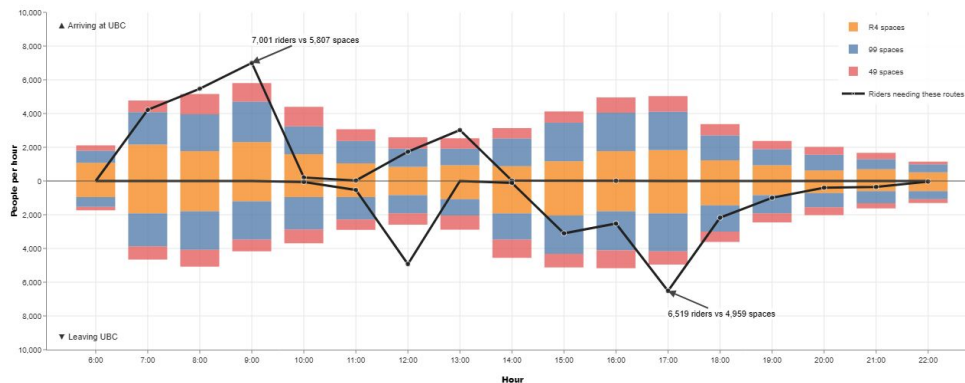


**Let's take a
closer look**

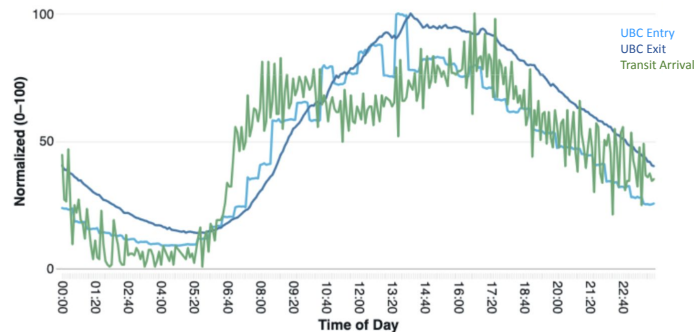
Rush Hour Exceeds Capacity



Rider demand vs. capacity on the R4, 99, and 49 (average weekday, term)



UBC: Entry/Exit vs Transit Arrivals (5-min)



7,001

riders arriving at 9 AM, room for 5,807

6,519

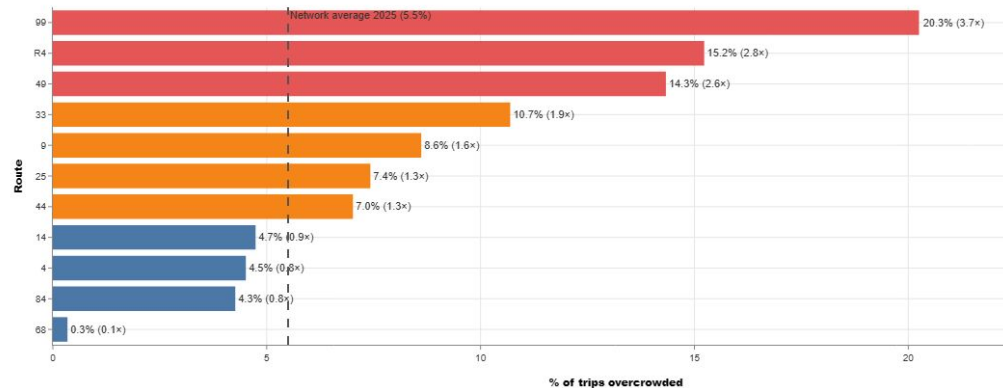
riders leaving at 5 PM, room for 4,959

The 5 PM shortfall holds under **every assumption** we tested.

Overcrowding by Route



Overcrowding on UBC bus routes (TransLink TSPR 2025)



20.3%

of 99 B-Line trips overcrowded

3.7x

the 5.5% network average

R4 · 99 · 49

the three most crowded UBC routes

Crowding directly due to UBC students

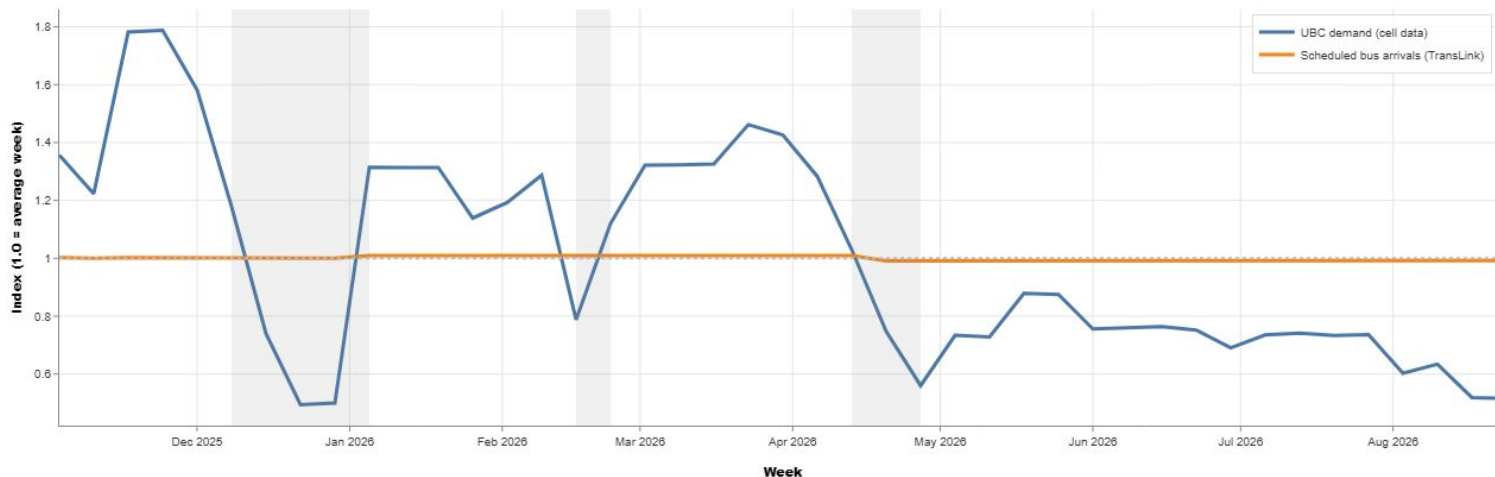


Source: TransLink Transit Service Performance Review 2025, route-level open data, and UBC open timetable data. UBC student crowding was estimated using a Monte Carlo simulation of students in transit, calibrated with cell tower data.

Lack of Flexibility & Adaptation



UBC demand vs. scheduled bus service



3.6×

swing in weekly campus demand

±2.5%

change in weekday buses to UBC

Dec 29

full weekday service in winter break

The Overall Problem

Overcrowded Corridors

- 20.3% of 99 trips and 15.2% of R4 trips are overcrowded.
- About 3–4× the 5.5% network average.

Lack of Flexibility & Adaptation

- Campus demand swings 3.6× across the year.
- Weekday buses to UBC change by only ±2.5%.
- Full service even during winter break.

Rush Hour Exceeds Capacity

- At 5 PM, 6,500 riders leave UBC on the R4, 99 and 49.
- Room for only 5,000, even at maximum load.

Meet Priya - UBC Student



Background

Program: 3rd-year Science

Lives in: Burnaby

Commute: Expo Line + R4, ~70 min

Classes: often end at 4:50 PM

Pass: U-Pass, rides transit daily

What matters to her

Getting on the first bus



Predictable travel time



Affordable transit



Pain Points

Watches 2-3 full R4s leave

Can't plan around the queue

Late for evening commitments

Often tired and frustrated

Meet Jacob - Service Planner



Background

Role: Service Planner, TransLink

Plans: bus routes serving UBC

Schedules: changed at each sign-up period

Data today: annual reports, rider complaints

Limits: no new buses (fleet and depots)

What matters to him

Relieving overcrowding



Staying within budget



Fairness to all riders



Pain Points

Crowding data comes as seasonal averages

No week-by-week view of campus demand

Can't add buses when routes are full

Every change affects another route's riders



Introducing to you our solution

termsync

What is it?

A planning tool that tells TransLink when campus demand will outgrow the schedule, and which existing buses can be moved to meet it, without buying new ones.

Detect



Flags the **weeks** when demand will exceed capacity, using campus data and the academic calendar.

Recommend



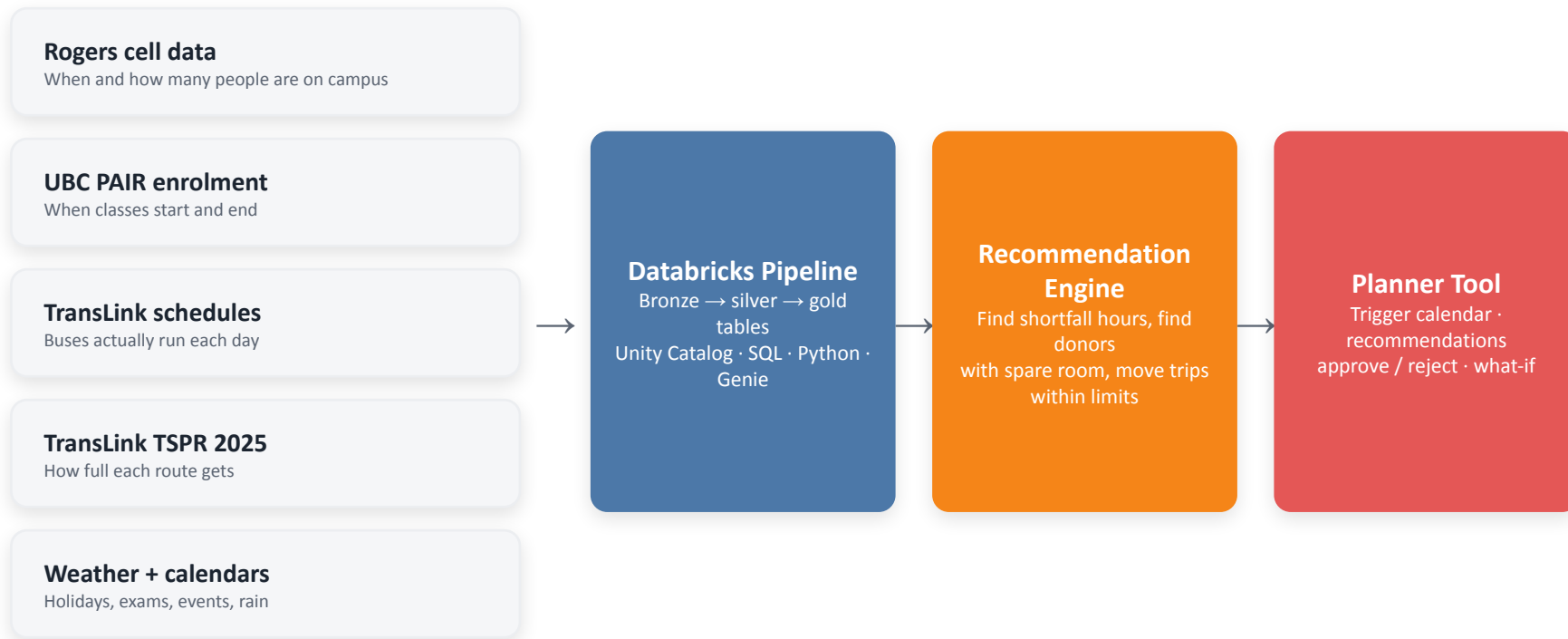
Moves trips from **quieter routes** that share UBC Exchange to the crowded corridor.

Protect

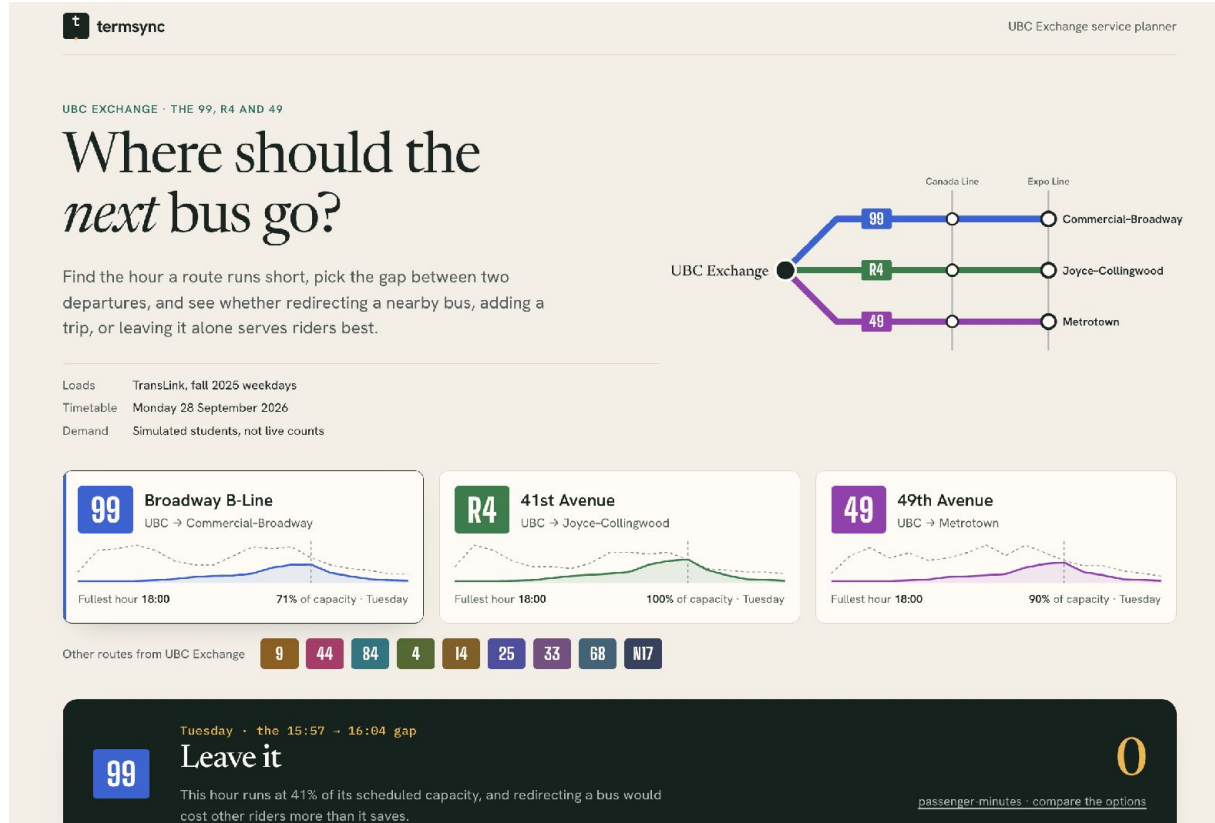


Donor routes stay **≤85% full** with a bus at least every **15 minutes**.

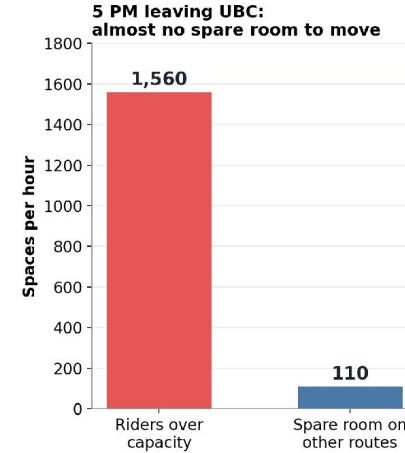
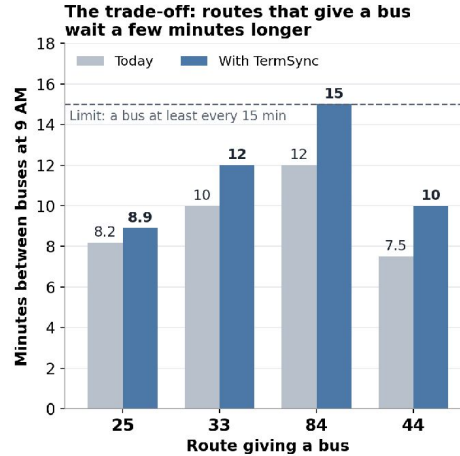
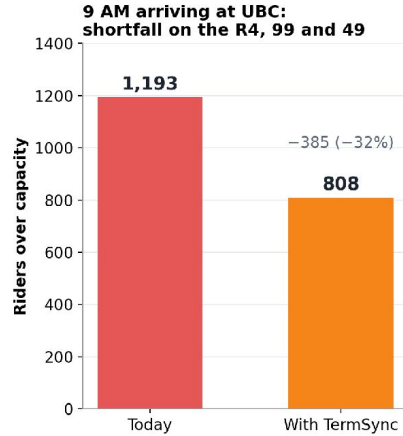
How We Built It



The Tool in Action



Impact & Value



-32%

of the 9 AM shortfall closed with existing buses

~8 buses

of peak capacity moved to crowded routes

\$ 9M

in vehicles TransLink doesn't need to buy

SWOT ANALYSIS

S

Strengths

- Uses buses already running
- Week-by-week demand TransLink lacks
- Donor limits built in

W

Weaknesses

- Synthetic cell data
- 55% corridor share is an estimate
- Hourly totals hide surges

O

Opportunities

- Broadway Subway changes in 2027
- Other campuses (SFU, Langara)

T

Threats

- Fleet and depot limits
- Evening peak full everywhere
- Operator shift rules

THANK YOU!

Questions?

Appendix: Methodology



Demand timing

Monte Carlo simulation of student schedules based on UBC PAIR enrolment by half hour, with transit destinations based on cell tower origin data.

Demand volume

Scaled to UBC's 79,000 weekday transit trips (2024). 55% assumed on the R4, 99 and 49.

Capacity

Scheduled trips at UBC Exchange, term weekdays, from the TransLink schedule in effect each day $\times$ max load (120 / 77).

Donor spare room

TSPR 2025 peak load factors. Donors give trips only while $\leq 85\%$ full and ≥ 4 trips per hour.

Appendix: Sensitivity



Share on R4, 99, 49	Arriving over capacity	Leaving over capacity
43% (R4 + 99 only)	None	12:00 (+1,261) · 17:00 (+137)
55% (estimate)	8:00 (+320) · 9:00 (+1,193) · 13:00 (+487)	12:00 (+2,337) · 17:00 (+1,560)
65% (upper bound)	7:00 (+217) · 8:00 (+1,315) · 9:00 (+2,466) · 13:00 (+1,037)	12:00 (+3,234) · 17:00 (+2,745)